

Playfair Cipher – Encryption & Decryption (Python)

```
def generate_matrix(key):
    key = key.upper().replace("J", "I")
    matrix = []
    used = set()

    for char in key:
        if char.isalpha() and char not in used:
            used.add(char)
            matrix.append(char)

    for char in "ABCDEFGHIKLMNOPQRSTUVWXYZ":
        if char not in used:
            used.add(char)
            matrix.append(char)

    return [matrix[i:i+5] for i in range(0, 25, 5)]

def find_position(matrix, char):
    if char == 'J':
        char = 'I'

    for i in range(5):
        for j in range(5):
            if matrix[i][j] == char:
                return i, j

def prepare_text(text):
    text = text.upper().replace("J", "I")
    text = ''.join(filter(str.isalpha, text))

    result = ""
    i = 0

    while i < len(text):
        a = text[i]

        if i + 1 < len(text):
            b = text[i + 1]
            if a == b:
                result += a + 'X'
                i += 1
            else:
                result += a + b
                i += 2
        else:
            result += a + 'X'
            i += 1

    return result

def playfair_encrypt(plaintext, key):
    matrix = generate_matrix(key)
    plaintext = prepare_text(plaintext)

    ciphertext = ""

    for i in range(0, len(plaintext), 2):
        a, b = plaintext[i], plaintext[i+1]

        row1, col1 = find_position(matrix, a)
        row2, col2 = find_position(matrix, b)

        if row1 == row2:
            ciphertext += matrix[row1][(col1 + 1) % 5]
            ciphertext += matrix[row2][(col2 + 1) % 5]

        elif col1 == col2:
            ciphertext += matrix[(row1 + 1) % 5][col1]
            ciphertext += matrix[(row2 + 1) % 5][col2]

        else:
            ciphertext += matrix[row1][col2]
            ciphertext += matrix[row2][col1]

    return ciphertext

def playfair_decrypt(ciphertext, key):
    matrix = generate_matrix(key)

    plaintext = ""

    for i in range(0, len(ciphertext), 2):
        a, b = ciphertext[i], ciphertext[i+1]

        row1, col1 = find_position(matrix, a)
```

```
        row2, col2 = find_position(matrix, b)

        if row1 == row2:
            plaintext += matrix[row1][(col1 - 1) % 5]
            plaintext += matrix[row2][(col2 - 1) % 5]

        elif col1 == col2:
            plaintext += matrix[(row1 - 1) % 5][col1]
            plaintext += matrix[(row2 - 1) % 5][col2]

        else:
            plaintext += matrix[row1][col2]
            plaintext += matrix[row2][col1]

    return plaintext

# Example
key = "MONARCHY"
plaintext = "INSTRUMENTS"

cipher = playfair_encrypt(plaintext, key)
print("Encrypted:", cipher)

decrypted = playfair_decrypt(cipher, key)
print("Decrypted:", decrypted)
```

Vigenère Cipher – Encryption & Decryption (Python)

```
def vigenere_encrypt(plaintext, key):
    plaintext = plaintext.upper()
    key = key.upper()

    ciphertext = ""
    key_index = 0

    for char in plaintext:
        if char.isalpha():
            p = ord(char) - ord('A')
            k = ord(key[key_index % len(key)]) - ord('A')

            c = (p + k) % 26
            ciphertext += chr(c + ord('A'))

            key_index += 1
        else:
            ciphertext += char

    return ciphertext

def vigenere_decrypt(ciphertext, key):
    ciphertext = ciphertext.upper()
    key = key.upper()

    plaintext = ""
    key_index = 0

    for char in ciphertext:
        if char.isalpha():
            c = ord(char) - ord('A')
            k = ord(key[key_index % len(key)]) - ord('A')

            p = (c - k + 26) % 26
            plaintext += chr(p + ord('A'))

            key_index += 1
        else:
            plaintext += char

    return plaintext

# Example
plaintext = "CYBERSECURITY"
key = "BEST"

cipher = vigenere_encrypt(plaintext, key)
print("Encrypted:", cipher)

decrypted = vigenere_decrypt(cipher, key)
print("Decrypted:", decrypted)
```